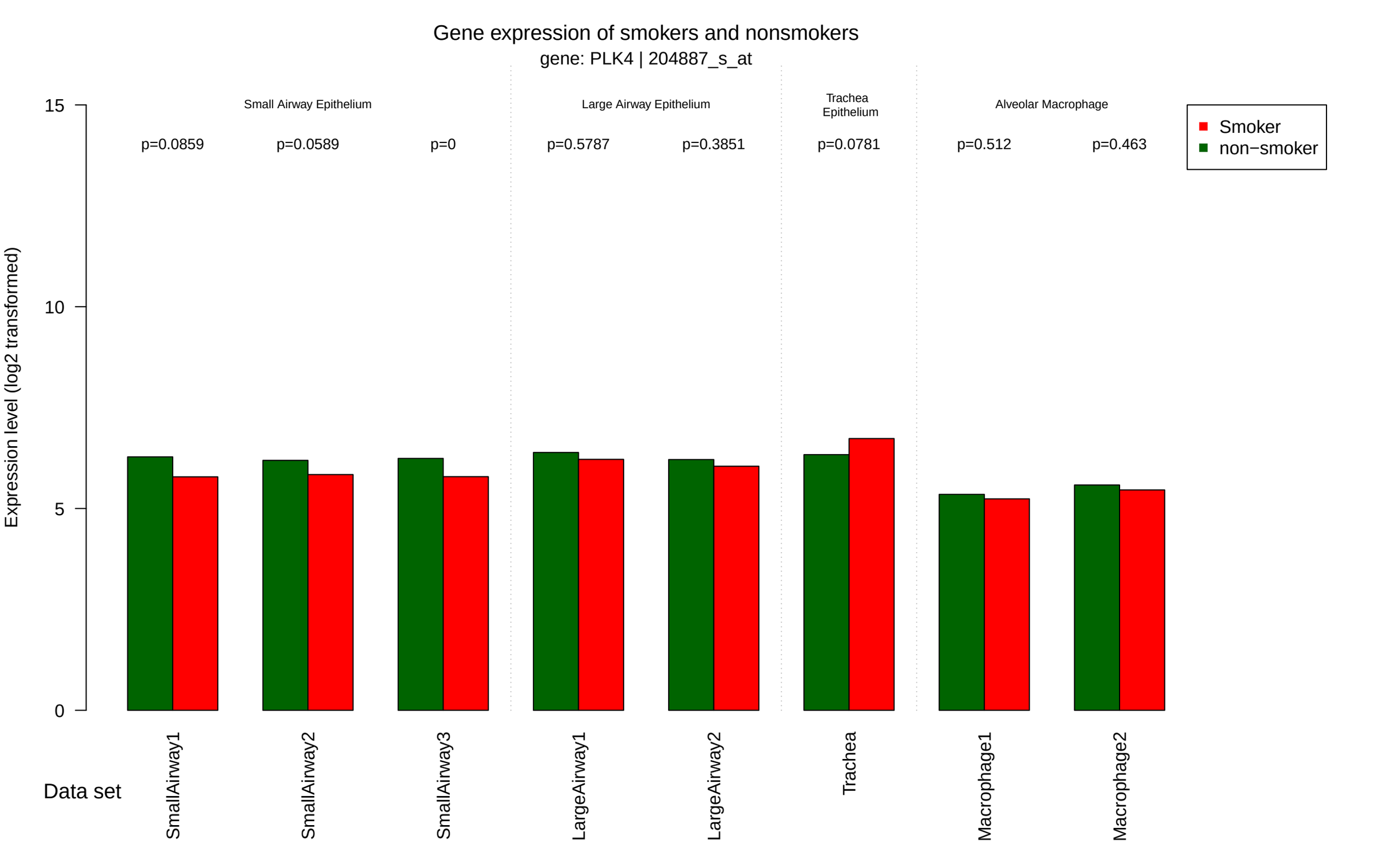
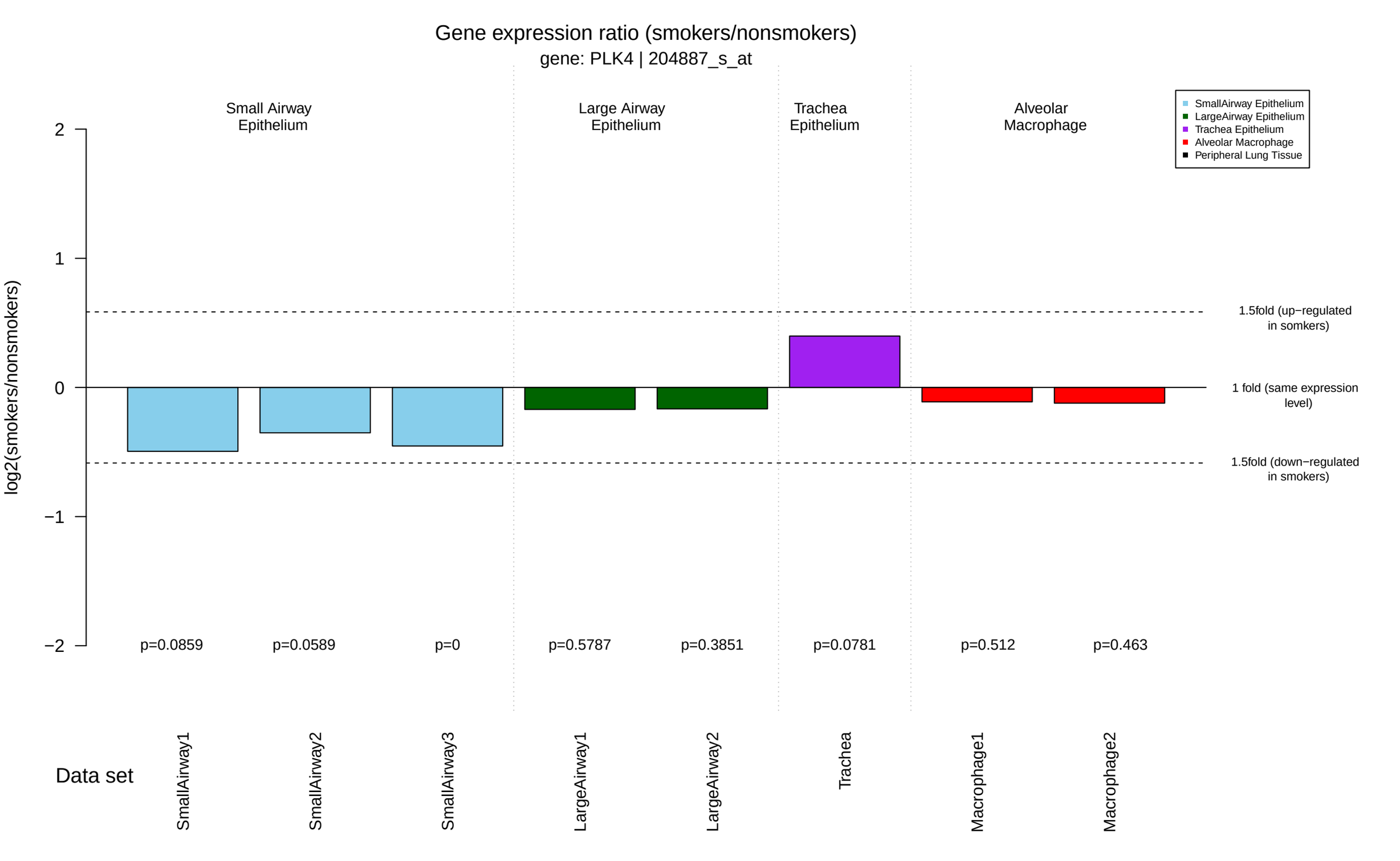
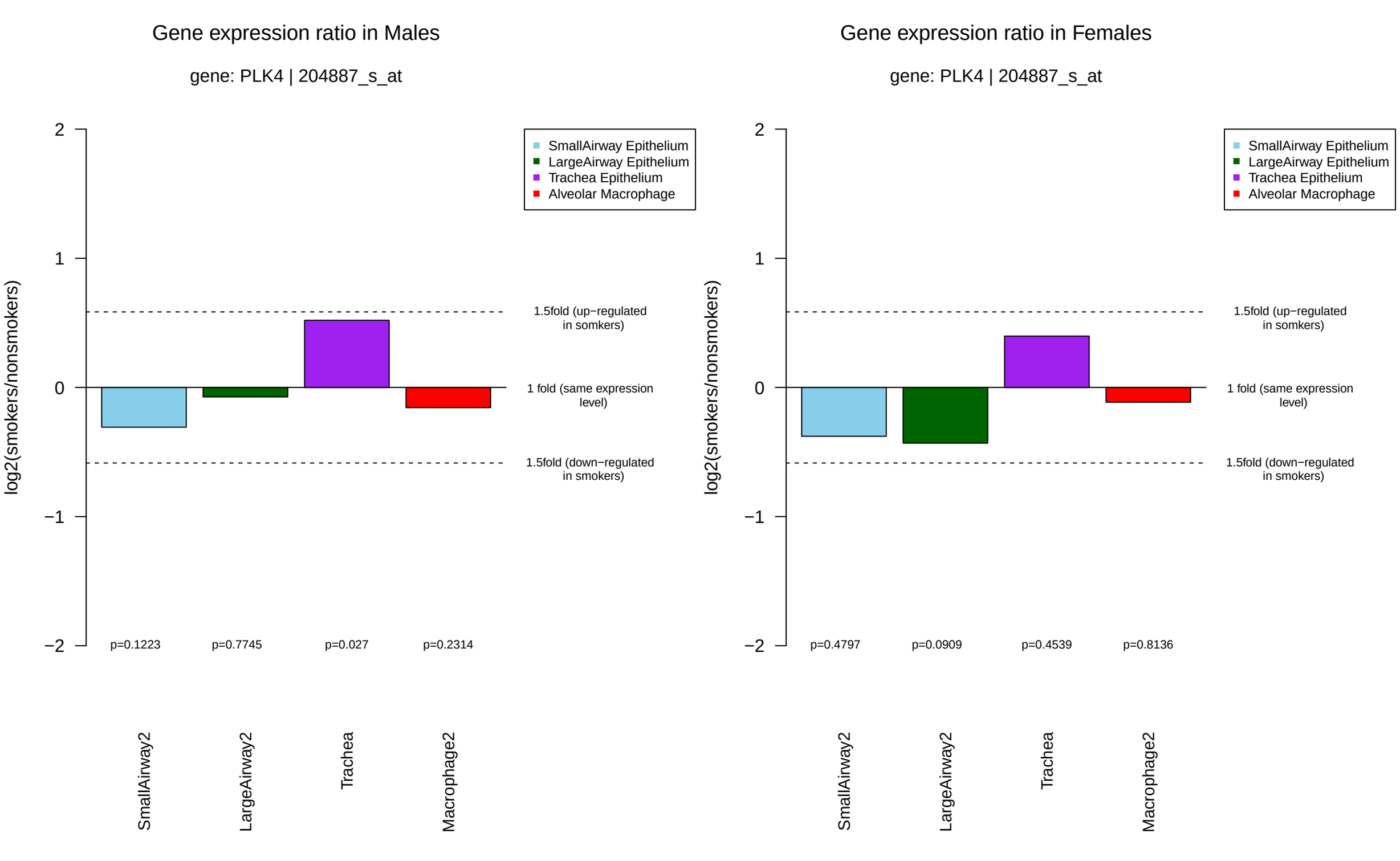
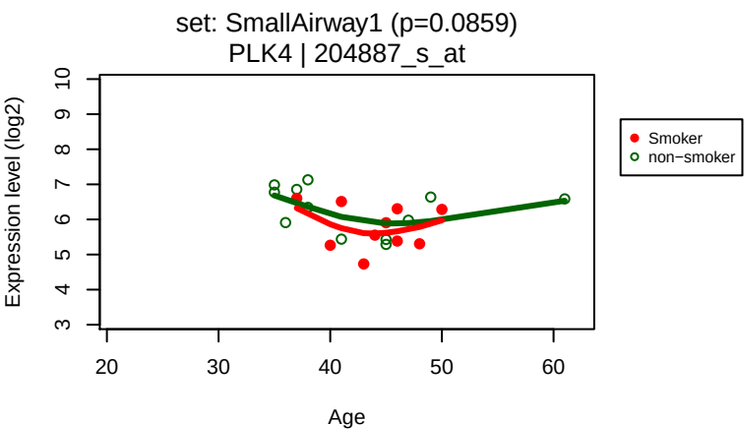
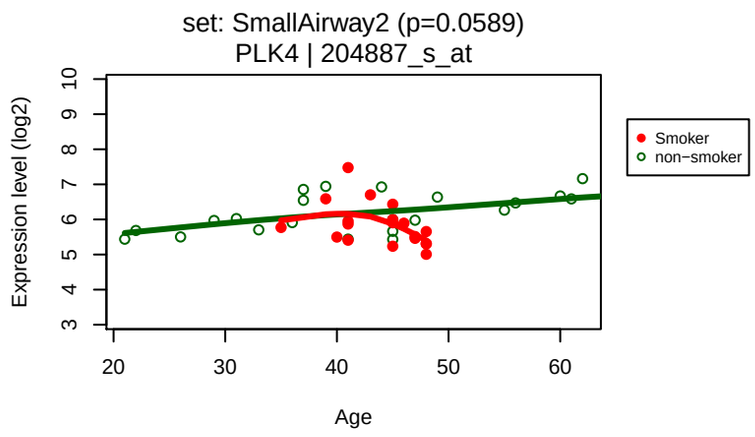
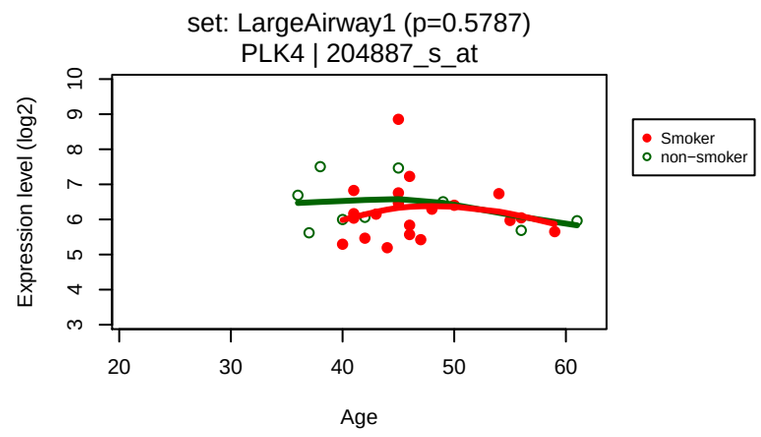

Gene expression of smokers and nonsmokers
gene: PLK4 | 204887_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0859
p=0.0589
p=0
p=0.5787
p=0.3851
p=0.0781
p=0.512
p=0.463
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

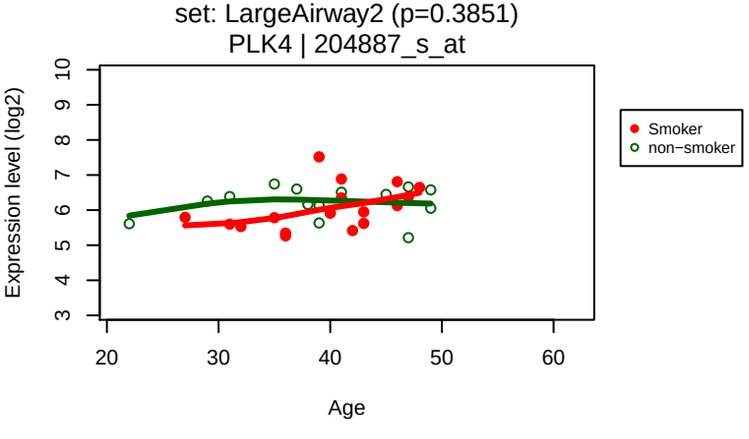

Gene expression ratio (smokers/nonsmokers)
gene: PLK4 | 204887_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0859
p=0.0589
p=0
p=0.5787
p=0.3851
p=0.0781
p=0.512
p=0.463
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

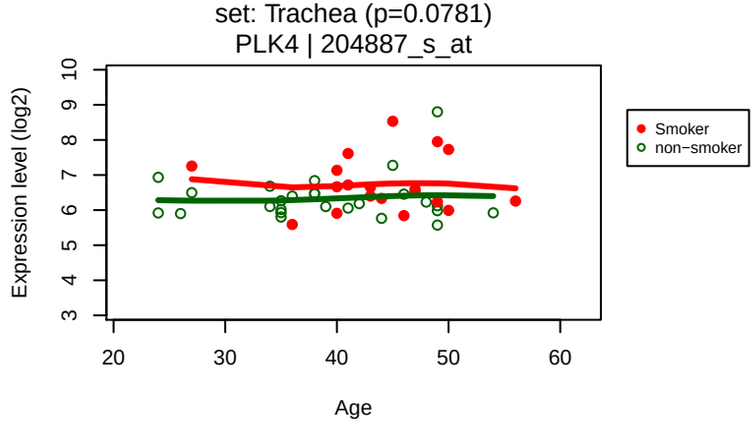

Gene expression ratio in Males
gene: PLK4 | 204887_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1223
p=0.7745
p=0.027
p=0.2314
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PLK4 | 204887_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4797
p=0.0909
p=0.4539
p=0.8136
SmallAirway2
LargeAirway2
Trachea
Macrophage2

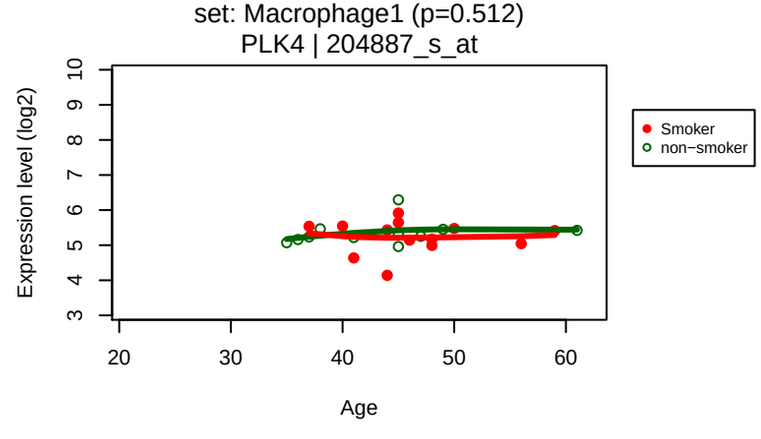

set: SmallAirway1 (p=0.0859)
PLK4 | 204887_s_at
Expression level (log2)
Age
Smoker
non−smoker

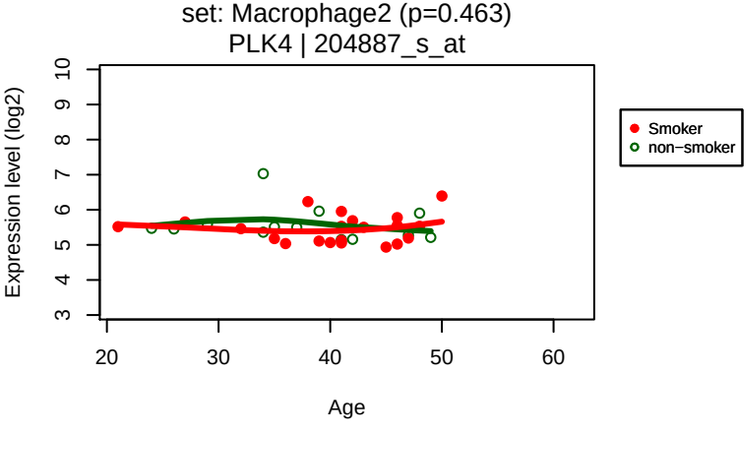

set: SmallAirway2 (p=0.0589)
PLK4 | 204887_s_at
Expression level (log2)
Age
Smoker
non−smoker

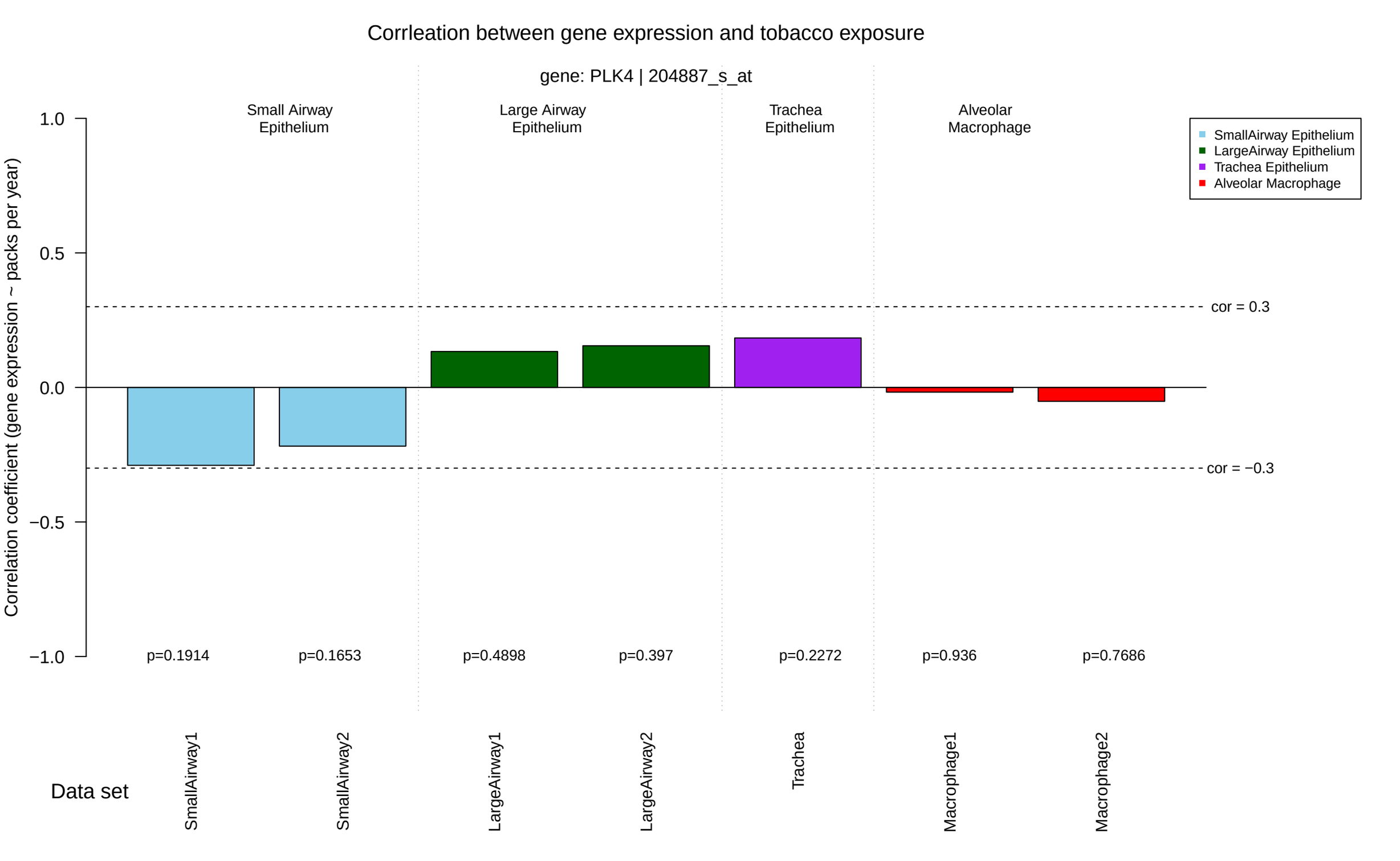

set: LargeAirway1 (p=0.5787)
PLK4 | 204887_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3851)
PLK4 | 204887_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0781)
PLK4 | 204887_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.512)
PLK4 | 204887_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.463)
PLK4 | 204887_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PLK4 | 204887_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1914
p=0.1653
p=0.4898
p=0.397
p=0.2272
p=0.936
p=0.7686
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2